

**JAWAHAR NAVODAYA VIDYALAYA
KOLLAM
(Hyderabad Region)**



SCIENCE EXHIBIT

GREEN PULSE

**(MODERN AGRICULTURAL
MONITORING AND
MANAGEMENT)**

CERTIFICATE

This is to certify that the project titled “**MODERN AGRICULTURAL MONITORING AND MANAGEMENT**” under the Science category has been done by Master

JAYAKRISHNAN S of class **XI**, **JAWAHAR NAVODAYA VIDYALAYA, KOLLAM** under my guidance and supervision for competing in the National level Science exhibition

2024-25

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Lastly, I want to thank my family for their constant support and belief in my abilities. Their encouragement has been my driving force in pursuing my interests and academic endeavours.

I am truly blessed to have such a supportive and inspiring community around me, and I am determined to continue learning and growing in the field of computer science. Thank you all once again for being an integral part of my journey

Sincerely,

JAYAKRISHNAN S

XI SCIENCE

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INTRODUCTION

In the modern era, agricultural efficiency and productivity are paramount for meeting the global food demand. "Green Pulse" is an innovative project aimed at transforming traditional farming practices through technology. This project leverages the capabilities of the Arduino Uno microcontroller, coupled with various sensors and modules, to create a comprehensive agricultural field monitoring and management system. Green Pulse is designed to enhance field security, monitor environmental conditions, and automate irrigation, thereby aiding farmers in optimizing their agricultural practices.

KEY FEATURES OF THE SYSTEM

1. Laser-Based Boundary Security with Buzzer Alert

- The project integrates a laser and Light Dependent Resistor (LDR) to create a virtual boundary around the field. When an object or person crosses the laser boundary, the LDR detects the interruption, triggering a buzzer alarm to alert the farmer of potential intrusions.

2. Radar for Bird Monitoring and Obstacle Avoidance

- An ultrasonic sensor, paired with a servo motor, functions as a radar system to detect birds and other above-ground obstacles.
- The sensor scans the area, and upon detecting an obstacle, it provides real-time alerts to prevent damage to crops.

3. Climate Monitoring with Temperature and Humidity Sensor

- A temperature and humidity sensor is included to monitor the climatic conditions of the field.
- This data helps farmers make informed decisions regarding crop management and protection against adverse weather conditions.

4. Automatic Irrigation System with Soil Moisture Sensor

- The soil moisture sensor measures the moisture level in the soil.
- When the moisture level drops below a predefined threshold, the relay module activates the water pump to irrigate the field automatically.
- This ensures optimal soil moisture levels, promoting healthy crop growth and conserving water.

5. Real-Time Data Display with LCD

- An LCD display is used to show real-time data, including temperature, humidity, and soil moisture levels.
- This feature provides farmers with immediate access to critical information, enhancing their ability to manage the effectively.

MATERIALS REQUIRED

NO	NAME OF THE COMPONENT	QTY
1	ARDUINO UNO MICROCONTROLLER	4
2	ARDUINO MEGA MICROCONTROLLER	1
3	PIR SENSOR	1
4	SERVO MOTOR	1
5	BUZZER	1
6	LIGHT DEPENDENT RESISTOR (LDR)	1
7	LASER MODULE	1
8	ULTRA SONIC SENSOR	1
9	TEMPERATURE-HUMIDITY SENSOR	1
10	RELAY MODULE	1
11	SOIL MOISTURE SENSOR	1
12	WATER PUMP	1
13	GRASS PATCHES	-
14	PINE WOOD	4
15	LCD MODULE 16 X 2	1
16	GSM MODULE WITH SIM CARD	1

SCIENTIFIC PRINCIPLE INVOLVED

1. Electromagnetic Radiation and Light Detection

- **Laser-Based Boundary Security:** The project uses a laser to create a boundary around the field. Lasers operate on the principle of electromagnetic radiation, emitting light in a narrow, concentrated beam. The Light Dependent Resistor (LDR) detects changes in light intensity. When the laser beam is interrupted, the LDR's resistance changes, triggering the buzzer. This principle is fundamental in optical communication and sensing technologies.

2. Ultrasonic Sound Waves

- **Radar for Bird Monitoring and Obstacle Avoidance:** Ultrasonic sensors emit high-frequency sound waves, which reflect off objects and return to the sensor. By measuring the time taken for the echoes to return, the sensor calculates the distance to the object. This principle, known as echolocation, is used in various applications, including medical imaging (ultrasound) and navigation systems.

3. Thermodynamics and Hygrometry

- **Temperature and Humidity Monitoring:**

Temperature sensors operate on the principle of thermodynamics, where changes in temperature affect the electrical properties of materials (such as resistance or voltage). Humidity sensors often use capacitive or resistive methods to measure moisture content in the air. These principles are crucial in meteorology, HVAC systems, and environmental monitoring.

4. Soil Science and Water Conservation

- **Soil Moisture Sensing and Automatic Irrigation:**

Soil moisture sensors typically use electrical resistance or capacitance to measure the water content in soil. As moisture levels change, the electrical properties of the soil alter, which the sensor detects. This principle is vital in soil science, aiding in efficient water management and conservation practices.

5. Electronics and Automation

- **Relay Module and Water Pump Control:** Relays are electronically operated switches that use an electromagnet to mechanically operate a switch, allowing low-power circuits to control high-power devices. In the Green Pulse project, the relay module controls the water pump based on the input from the soil moisture sensor. This automation principle is widely used in industrial control systems, home automation, and robotics.

CONSTRUCTION AND WORKING

The **Laser-Based Boundary System** uses a laser, mirrors, and an LDR to create a reflective path. If an intruder interrupts the laser beam, the LDR triggers a buzzer and sends an SMS alert to the farmer, enhancing security against animal intrusions. The **Radar for Bird Monitoring** employs an ultrasonic sensor mounted on a rotating servo motor to detect birds or obstacles by measuring reflected sound waves, alerting the farmer to potential threats. The **Burrow Detection System** identifies underground burrow activity using sensors placed strategically to alert the farmer of burrowing animals that might harm crops or roots. The **Climate Monitoring System** uses a temperature and humidity sensor to provide real-time data on an LCD display, helping farmers make climate-related decisions. Finally, the **Automatic Irrigation System** includes a soil moisture sensor that activates a relay to control a water pump, automatically irrigating the field when moisture levels are low, thus conserving water and ensuring optimal soil moisture for crops.

Here's a basic outline of how the Green Pulse works with arduino, and other sensors :

1. Laser-Based Boundary System

- **Components:** Single laser, Light Dependent Resistor (LDR), mirrors, GSM module for SMS alert.
- **Setup:** The laser is positioned at one corner of the field, while mirrors are strategically placed at the remaining three corners to create a continuous reflective path. The LDR is positioned back at the starting point, aligned to detect the laser beam after reflection.
- **Principle:** The system operates based on the principle of reflection. The laser beam is directed through mirrors at each corner, forming an uninterrupted path back to the LDR. When an intruder disrupts this beam, the LDR's resistance changes,

which triggers both a buzzer alarm and an SMS alert to notify the farmer of the intrusion.

2. Radar for Bird Monitoring and Obstacle Avoidance

- **Components:** Ultrasonic sensor, servo motor.
- **Setup:** The ultrasonic sensor is mounted on a servo motor, which allows it to rotate and scan the field.
- **Principle:** The ultrasonic sensor emits sound waves that bounce off objects and return. By measuring the time taken for the echoes to return, the distance to obstacles is calculated, and real-time alerts are generated.

3. Climate Monitoring System

- **Components:** Temperature and humidity sensor, LCD display.
- **Setup:** The temperature and humidity sensor is placed in the field to monitor environmental conditions. Data is sent to the Arduino Uno and displayed on the LCD.
- **Principle:** The sensor measures temperature and humidity levels, and the LCD provides real-time data visualization for the farmer.

4. Automatic Irrigation System

- **Components:** Soil moisture sensor, relay module, water pump.
- **Setup:** The soil moisture sensor is inserted into the soil to monitor moisture levels. The relay module controls the water pump based on the sensor's readings.
- **Principle:** When soil moisture drops below a threshold, the sensor signals the Arduino to activate the relay, turning on the water pump to irrigate the field.

5. Burrow Detection System

- **Components:** Burrow-specific sensors, microcontroller, buzzer.

- **Setup:** Sensors are embedded slightly below ground level in areas prone to burrowing. These sensors are positioned to detect ground disturbances or the presence of underground activity near the crop roots.
- **Principle:** The system operates by detecting soil vibrations or movements caused by burrowing animals. When these disturbances are sensed, the system registers a change, which triggers an alert through the buzzer. This setup allows for early warning of burrowing activity that could harm crops or roots.

Working Mechanism

1. Laser-Based Boundary System

- The laser beam creates a continuous path via mirrors around the field.
- When the laser beam is uninterrupted, the LDR detects it, indicating normal conditions.
- If the beam is interrupted, the LDR's resistance changes, triggering the buzzer and send SMS to alert the farmer of potential intrusions.

2. Radar for Bird Monitoring and Obstacle Avoidance

- The ultrasonic sensor continuously emits sound waves and measures the time taken for echoes to return.
- Mounted on a servo motor, it rotates to scan the field.
- Detected obstacles or birds generate alerts, helping prevent crop damage.

3. Climate Monitoring System

- The temperature and humidity sensor collects data on environmental conditions.
- The Arduino processes this data and displays it on the LCD.
- Farmers can monitor real-time climate data to make informed decisions.

4. Automatic Irrigation System

- The soil moisture sensor continuously measures the

soil's moisture level.

- When moisture levels are low, the sensor signals the Arduino to activate the relay.
- The relay turns on the water pump, automatically irrigating the field until optimal moisture levels are restored.

5.Burrow Detection System

- The sensors are embedded near the ground's surface in areas where burrowing activity is likely.
- When no underground movement is detected, the sensors remain inactive, indicating normal conditions.
- If soil disturbances or vibrations from burrowing animals are detected, the sensors register a change, triggering the buzzer to alert the farmer of potential crop or root damage from underground activity.

By integrating these components, Green Pulse effectively monitors field boundaries, detects obstacles, tracks environmental conditions, and automates irrigation processes.

This system enhances agricultural efficiency, supports sustainable practices, and helps farmers manage their fields more effectively. Additionally, consider implementing error handling and optimizing the code for better performance. Always follow safety guidelines while using sensors and microcontrollers

In summary, The system features a laser boundary with mirrors and an LDR to detect intrusions via beam interruption, an ultrasonic sensor mounted on a servo motor for radar-based bird and obstacle detection, and a temperature and humidity sensor for climate monitoring. Data from these sensors is displayed on an LCD for real-time insights. Additionally, a soil moisture sensor triggers an automatic irrigation system through a relay-controlled water pump, ensuring efficient water use and maintaining optimal soil conditions. Green Pulse integrates these components to offer comprehensive, real-time field management and automation

ADVANTAGES OF THE PROJECT

1. Enhanced Field Security

- **Laser-Based Boundary:** The laser and LDR setup creates a virtual perimeter around the field, providing an effective way to detect unauthorized entries and protect crops from potential damage or theft.

2. Improved Crop Protection

- **Radar System:** The ultrasonic sensor's radar capability helps monitor and avoid birds and other above-ground obstacles, reducing the risk of crop damage and ensuring healthier plants.

3. Optimized Resource Management

- **Automatic Irrigation:** The soil moisture sensor, combined with the relay-controlled water pump, ensures that crops receive the right amount of water, conserving water resources and preventing over- or under-watering.

4. Real-Time Climate Monitoring

- **Temperature and Humidity Sensors:** Continuous monitoring of environmental conditions allows farmers to make informed decisions based on current climate data, improving crop management and adaptation to weather changes.

5. Efficient Data Visualization

- **LCD Display:** Real-time data on temperature, humidity, and soil moisture is readily accessible via the LCD, enabling quick assessments and timely responses to field conditions.

6. Automated System Integration

- **Seamless Operation:** The integration of sensors and modules with the Arduino Uno allows for automated and coordinated control of field

management tasks, reducing the need for manual intervention and increasing operational efficiency.

7. Enhanced Agricultural Productivity

- **Comprehensive Monitoring:** By addressing security, climate, and irrigation needs, Green Pulse helps improve overall crop health and yield, contributing to more productive and sustainable farming practices.

8. Resource Conservation

- **Efficient Irrigation:** Automated irrigation based on real-time soil moisture levels helps in conserving water and ensuring that resources are used effectively, aligning with sustainable agriculture goals.

DISADVANTAGES OF MY PROJECT

- **Installation:** Setting up the laser-based boundary system, ultrasonic sensor, and other components requires precise alignment and calibration, which can be complex and time-consuming.
- **Sensor Limitations:** The ultrasonic sensor's effective range and the laser boundary's coverage might be limited, affecting their ability to monitor large or irregularly shaped fields comprehensively

CONCLUSION

In conclusion, Green Pulse represents a significant advancement in agricultural field monitoring and management, offering a comprehensive solution for optimizing farm operations through technology. By integrating a variety of sensors and modules with an Arduino Uno microcontroller, the project provides a multifaceted approach to addressing key challenges in modern agriculture. The primary advantage of Green Pulse lies in its ability to enhance field security through a laser-based boundary system and an ultrasonic radar setup. The laser and LDR combination creates a virtual perimeter around the field, providing a reliable means of detecting intrusions and safeguarding crops from potential threats.

Simultaneously, the ultrasonic sensor's radar functionality helps monitor and avoid birds and other above-ground obstacles, reducing the risk of crop damage and ensuring a healthier harvest.

Moreover, Green Pulse excels in optimizing resource management through its automatic irrigation system. The soil moisture sensor, coupled with a relay-controlled water pump, ensures that crops receive the appropriate amount of water based on real-time moisture levels. This not only conserves water but also promotes efficient irrigation practices, contributing to sustainable agriculture. The inclusion of temperature and humidity sensors further enhances the system's capabilities by providing real-time climate data. This information is displayed on an LCD,

enabling farmers to make informed decisions about crop management and adaptation to changing weather conditions. Such timely insights are crucial for maintaining optimal growing conditions and improving overall crop yield.

Despite its advantages, Green Pulse does present certain challenges. The complexity of installation and setup, the cost of components, and the need for regular maintenance are significant considerations. Additionally, environmental factors may impact the performance of the laser and ultrasonic sensors, and power consumption could pose a concern for some users. The system's scalability for larger or irregularly shaped fields also requires careful planning and potential adjustments. GreenPulse offers a robust and innovative solution for agricultural field monitoring, combining advanced technology with practical applications. While it may require a considerable initial investment and ongoing maintenance, the benefits of improved field security, resource optimization, and real-time climate monitoring make it a valuable tool for modern farmers. By addressing both the opportunities and limitations of the system, Green Pulse paves the way for more efficient and sustainable agricultural practices, contributing to the overall goal of enhancing food production and environmental steward

